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ADVANCED ANALOG INTEGRATED CIRCUIT DESIGN

Design and Analysis of a Two-Stage Miller-Compensated CMOS Operational Amplifier

GPDK090, 1.2 V Supply

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Technology: GPDK090 BSIM3
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This report separates hand calculations, transistor-level operating-point checks, and simulation measurements. Values not directly demonstrated by the available plots are explicitly marked as estimates or re-test items.

Abstract

This report presents the hand design and transistor-level verification of an unbuffered, two-stage CMOS operational amplifier with Miller compensation. The circuit uses an NMOS differential input pair, a PMOS current-mirror active load, an NMOS tail source, a PMOS common-source second stage, and an NMOS sink load. The nominal supply is 1.2 V; the external load is 10 pF; the selected Miller capacitor is 3.6 pF; and the bias reference is 6.1 μA.

The design was initially sized for a target dc gain of at least 70 dB, a gain-bandwidth product of 5 MHz, a slew rate of 5 V/μs, an input common-mode range of 0.4–0.9 V, and approximately 60° phase margin. Spectre simulation gives 74.05 dB low-frequency open-loop gain, a 851.14 Hz dominant-pole bandwidth, a near-unity-gain frequency of approximately 3.72 MHz, and approximately 54.3° phase margin. The simulated 1% settling times are about 0.22 μs and 0.25 μs for positive and negative steps. The low-frequency CMRR, PSRR^+, and PSRR^- are approximately 79.62 dB, 85.75 dB, and 84.13 dB, respectively.

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Chapter 1

Introduction and Design Requirements

1.1 Objective

The objective is to design, size, and verify a compact two-stage CMOS operational amplifier in the GPDK090 process. The report follows a calculation-first flow: analytical equations establish a first-cut design, dc operating-point data checks bias and saturation, ac analysis evaluates gain and stability, and transient analysis evaluates large-signal speed and settling. The design method and small-signal interpretation follow the standard treatments of analog CMOS design in Razavi and Baker [1, 2], together with the unbuffered two-stage design procedure summarized by Allen [3].

1.2 Architecture

The transistor functions are:

- M1–M2: matched NMOS differential input pair;
- M3–M4: matched PMOS current-mirror load and single-ended conversion;
- M5: NMOS tail-current source;
- M6: PMOS common-source second gain stage;
- M7: NMOS output-stage current sink, implemented with multiplicity $m = 6$;
- M8: diode-connected NMOS device that converts I_{BIAS} into the bias voltage for M5 and M7;
- C_C: Miller compensation capacitor from the first-stage output to the amplifier output.

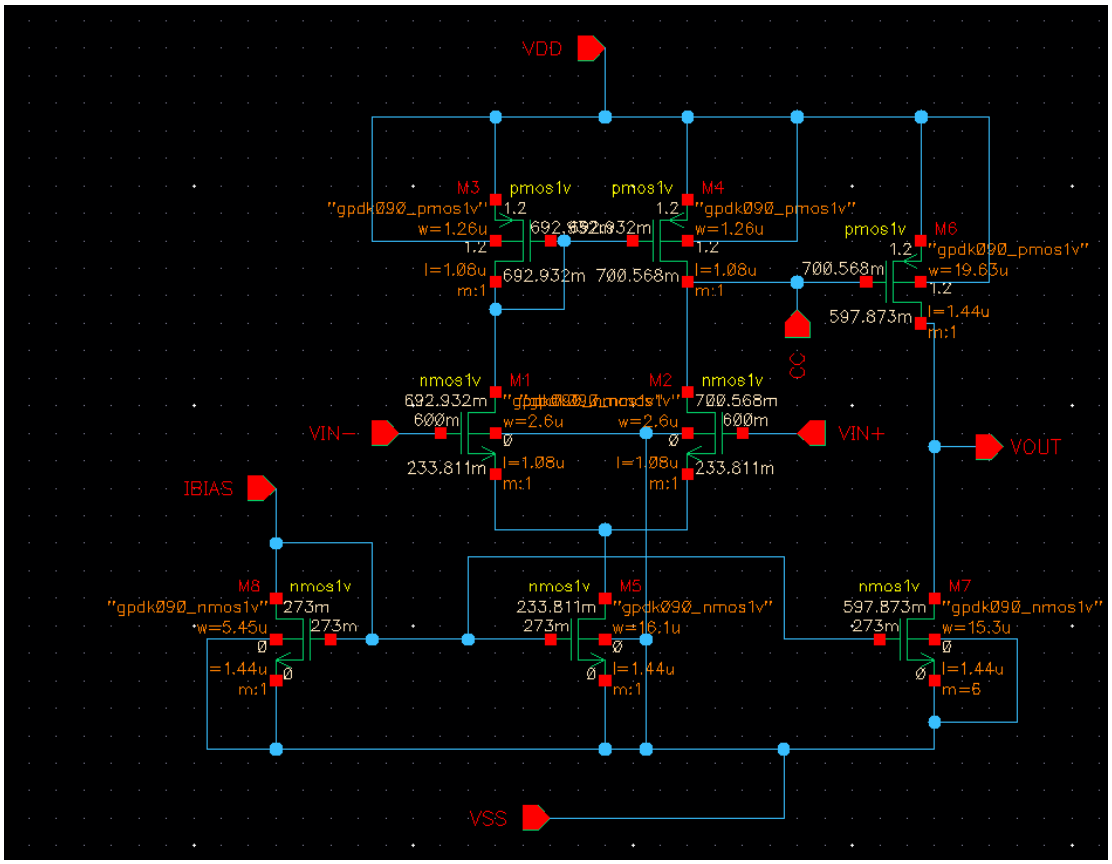


Figure 1.1: Final transistor-level schematic with dc node voltages and device dimensions.

1.3 Target specifications and design constants

Table 1.1: Design targets and fixed conditions.

Parameter	Target / selected value	Symbol
Supply rails	$V_{DD} = 1.2\,\text{V}$, $V_{SS} = 0\,\text{V}$	V_{DD}, V_{SS}
DC open-loop gain	at least 70 dB	A_0
Gain-bandwidth product	5 MHz	GBW
Phase margin	approximately 60° or higher	PM
Load capacitance	10 pF	C_L
Miller capacitance	3.6 pF	C_C
Slew rate	$5\,\text{V}/\mu\text{s}$	SR
1% settling time	at most $1\,\mu$s for a 0.4 V step	t_s
Input common-mode range	0.4–0.9 V	ICMR
Bias reference	$6.1\,\mu$A	I_{BIAS}
Process transconductance parameters	$K'_n = 296.4\,\mu\text{A/V}^2$, $K'_p = 167.4\,\mu\text{A/V}^2$	K'_n, K'_p

The threshold values used in the first-cut sizing are extracted from the design work: $|V_{TP}|_{max} = 0.149135\,\text{V}$, $V_{TN,min} = 0.151885\,\text{V}$, and $V_{TN,max} = 0.186604\,\text{V}$. These simplified values are used only for hand sizing; Spectre uses the complete BSIM3 model.

Chapter 2

First-Cut Analytical Design

2.1 Design relations

For $S_i = (W/L)_i$ and matched mirror devices, the principal relations are

$$SR \approx \frac{I_5}{C_C}, \tag{2.1}$$

$$GBW \approx \frac{g_{m1}}{2\pi C_C}, \tag{2.2}$$

$$A_{v1} \approx \frac{g_{m2}}{g_{ds2} + g_{ds4}}, \tag{2.3}$$

$$A_{v2} \approx \frac{g_{m6}}{g_{ds6} + g_{ds7}}, \tag{2.4}$$

$$A_0 \approx A_{v1} A_{v2}, \tag{2.5}$$

$$p_2 \approx -\frac{g_{m6}}{C_L}, \qquad z_{RHP} \approx \frac{g_{m6}}{C_C}. \tag{2.6}$$

For the classical Miller-compensated topology without a nulling resistor, a practical 60° phase-margin rule is

$$g_{m6} \approx 2.2\, g_{m2} \frac{C_L}{C_C}, \tag{2.7}$$

provided that other high-frequency poles are well above the unity-gain frequency.

2.2 Compensation capacitor and tail current

The minimum compensation capacitor from the design rule is

$$C_C > 0.22 C_L = 0.22(10\,\text{pF}) = 2.2\,\text{pF}. \tag{2.8}$$

The selected value $C_C = 3.6\,\text{pF}$ provides additional pole separation. From the slew-rate target,

$$I_5 = SR\, C_C = (5\,\text{V}/\mu\text{s})(3.6\,\text{pF}) = \boxed{18\,\mu\text{A}}. \tag{2.9}$$

2.3 Input pair and PMOS mirror

The required input-pair transconductance is

$$g_{m1} = 2\pi (GBW) C_C = 2\pi (5\,\text{MHz})(3.6\,\text{pF}) = \boxed{113.10\,\mu\text{S}}. \tag{2.10}$$

At $V_{in,max} = 0.9\,\mathrm{V}$, the mirror overdrive used for the first cut is

$$V_{OV3} = V_{DD} - V_{in,max} - |V_{TP}|_{max} + V_{TN,min} \tag{2.11}$$
$$= 1.2 - 0.9 - 0.149135 + 0.151885 = 0.30275\,\mathrm{V}. \tag{2.12}$$

Therefore,

$$S_3 = S_4 = \frac{I_5}{K'_p V_{OV3}^2} = \boxed{1.173}. \tag{2.13}$$

For M1 and M2,

$$S_1 = S_2 = \frac{g_{m1}^2}{K'_n I_5} = \boxed{2.397}. \tag{2.14}$$

2.4 Tail transistor M5

The input-pair overdrive from the first-cut values is

$$V_{OV1} = \sqrt{\frac{I_5}{K'_n S_1}} = 0.15915\,\mathrm{V}. \tag{2.15}$$

A conservative hand calculation used $V_{in,min} = 0.45\,\mathrm{V}$ to keep finite saturation margin:

$$V_{DS5,sat} = V_{in,min} - V_{SS} - V_{OV1} - V_{TN,max} \tag{2.16}$$
$$= 0.45 - 0.15915 - 0.186604 = 0.10424\,\mathrm{V}, \tag{2.17}$$
$$S_5 = \frac{2I_5}{K'_n V_{DS5,sat}^2} = \boxed{11.178}. \tag{2.18}$$

This conservative equation predicts a lower ICMR near 0.45 V. The requested 0.4 V lower limit must therefore be verified by the transistor model rather than assumed from the square-law estimate.

2.5 Second stage M6 and output sink M7

The PMOS mirror transconductance estimated from the first stage is

$$g_{m4} = \frac{I_5}{V_{OV3}} = 59.45\,\mu\mathrm{S}. \tag{2.19}$$

The transconductance required for approximate 60° phase margin is

$$g_{m6} = 2.2 g_{m2} \frac{C_L}{C_C} = 2.2(113.10\,\mu\mathrm{S})\frac{10}{3.6} = \boxed{691.15\,\mu\mathrm{S}}. \tag{2.20}$$

Using the matched-mirror relationship,

$$S_6 = \frac{g_{m6}}{g_{m4}} S_4 = \boxed{13.637}, \tag{2.21}$$

$$I_6 = \frac{g_{m6}^2}{2K'_p S_6} = \boxed{104.62\,\mu\mathrm{A}}, \tag{2.22}$$

$$S_7 = \frac{I_6}{I_5} S_5 = \boxed{64.968}. \tag{2.23}$$

2.6 Bias device M8

M8 and M5 share the same gate-source voltage. Hence their drain currents scale with geometry:

$$\frac{I_{BIAS}}{I_5} = \frac{S_8}{S_5} quad \Rightarrow \quad S_8 = S_5 \frac{I_{BIAS}}{I_5} = 11.178 \frac{6.1}{18} = \boxed{3.788}. \tag{2.24}$$

With $L_8 = 1.44\,\mu\mathrm{m}$, the calculated width is $W_8 = 5.45\,\mu\mathrm{m}$, which is the implemented value.

2.7 First-cut and implemented dimensions

Table 2.1: Calculated sizing and dimensions used in the final schematic.

Device	Calculated $S = W/L$	First-cut W (μm)	Final W (μm)	Final L (μm)	Multiplicity m
M1, M2	2.397	2.59	2.60	1.08	1
M3, M4	1.173	1.27	1.26	1.08	1
M5	11.178	16.10	16.10	1.44	1
M6	13.637	19.64	19.63	1.44	1
M7	64.968 effective	15.59 per finger	15.30	1.44	6
M8	3.788	5.45	5.45	1.44	1

The implemented dimensions preserve the calculated ratios while using practical channel lengths and multiplicity. M7 was rounded slightly during dc balancing. Such small tuning is expected because the square-law equations neglect channel-length modulation, body effect, mobility degradation, and intrinsic capacitances.

The idealized quiescent power estimate is

$$P_{DC} \approx V_{DD}(I_5 + I_6 + I_{BIAS}) = 1.2(18 + 104.62 + 6.1)\,\mu\text{A} = \boxed{154.5\,\mu\text{W}}. \tag{2.25}$$

This is an analytical estimate; the final report should use the Spectre supply-current measurement if an exact power value is required.

Chapter 3

DC Analysis

3.1 Operating-point verification

The dc operating point was evaluated at $V_{IN+} = V_{IN-} = 0.6\,\mathrm{V}$ with $C_L = 10\,\mathrm{pF}$. The schematic reports $V_{OUT} \approx 0.598\,\mathrm{V}$ and every device is marked in region 2, which is the simulator's saturation-region code for these MOS models.

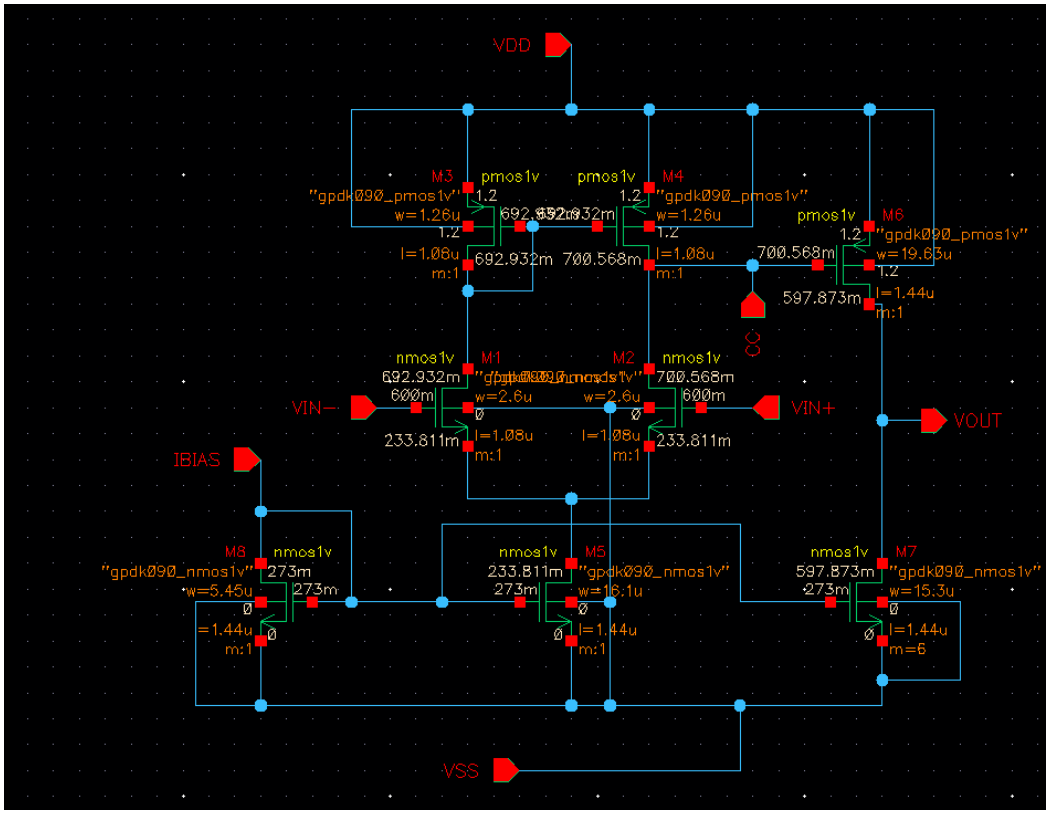


Figure 3.1: Annotated dc operating point at $V_{IN+} = V_{IN-} = 0.6\,\mathrm{V}$. The displayed node voltages make the headroom and stage balance directly visible.

The annotated schematic supports the following dc checks:

- The two input gates are both at 0.6 V, while the common-source node of M1–M2 is approximately 0.2338 V. The differential pair is therefore biased symmetrically.
- The two first-stage drain nodes are approximately 0.6929 V and 0.7006 V. Their small difference is consistent with the single-ended current-mirror conversion and device-model nonidealities.
- The M5 drain voltage is approximately 0.2338 V, which is greater than its recorded $V_{DS,sat} \approx 0.100$ V. M5 therefore retains about 0.134 V saturation margin.
- The bias node shared by M8, M5, and M7 is approximately 0.273 V. The close agreement between the intended 6.1 μA reference and the recorded M8 current confirms the bias-mirror scaling.
- The M6 gate voltage is approximately 0.7006 V, and $V_{OUT} = 0.5979$ V. Consequently, M6 has $V_{SD6} \approx 0.602$ V and M7 has $V_{DS7} \approx 0.598$ V, providing comfortable output-stage saturation headroom at the quiescent point.
- The output is only about 2.1 mV below the ideal mid-supply value of 0.6 V, showing that the final M6–M7 current balance is well centered.

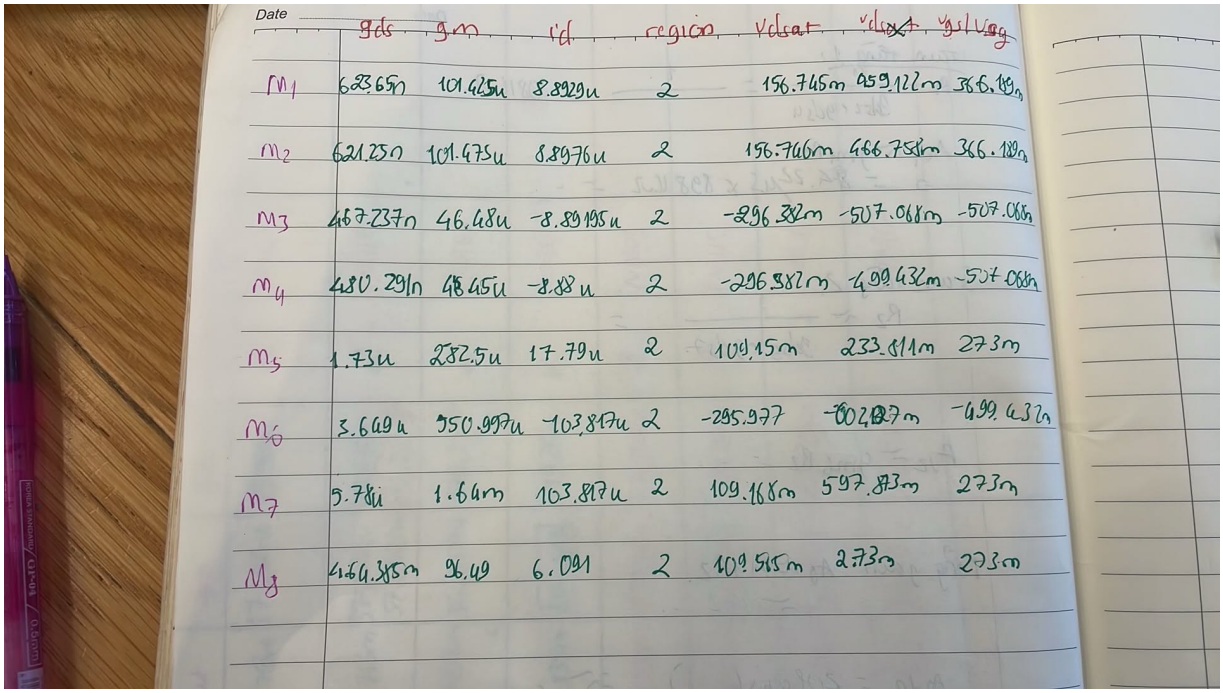


Figure 3.2: Recorded Spectre operating-point quantities for M1–M8.

The most useful quantities transcribed from the operating-point record are summarized below. Handwritten last digits are rounded; conclusions should rely on their order of magnitude rather than the final digit.

Table 3.1: Selected nominal operating-point data.

Device	I_D	g_m	g_{ds}	Region	$\lvert V_{DS}\rvert$
M1	8.893 µA	101.43 µS	6.26 nS	2	0.959 V
M2	8.898 µA	101.48 µS	6.21 nS	2	0.487 V
M3	8.892 µA	46.48 µS	467.24 nS	2	0.507 V
M4	8.83 µA	46.45 µS	480.29 nS	2	0.499 V
M5	17.79 µA	282.5 µS	1.73 µS	2	0.234 V
M6	103.82 µA	551.0 µS	3.65 µS	2	0.602 V
M7	103.82 µA	1.60 mS	5.78 µS	2	0.598 V
M8	6.09 µA	96.49 µS	0.464 µS	2	0.273 V

The current checks are consistent with the intended biasing:

$$I_{D1} + I_{D2} = 17.791\,\mu\text{A} \approx I_{D5} = 17.79\,\mu\text{A}, \tag{3.1}$$

and

$$I_{D6} \approx I_{D7} \approx 103.82\,\mu\text{A}. \tag{3.2}$$

3.2 Gain estimated from the dc operating point

Using the measured small-signal parameters,

$$A_{v1} \approx \frac{101.475\,\mu\text{S}}{6.21\,\text{nS} + 480.291\,\text{nS}} = 208.6 \quad (46.39\,\text{dB}), \tag{3.3}$$

$$A_{v2} \approx \frac{550.997\,\mu\text{S}}{3.649\,\mu\text{S} + 5.78\,\mu\text{S}} = 58.44 \quad (35.33\,\text{dB}), \tag{3.4}$$

$$A_{0,OP} \approx A_{v1}A_{v2} = 1.219 \times 10^4 \quad (81.72\,\text{dB}). \tag{3.5}$$

The full ac simulation gives 74.05 dB. The approximately 7.7 dB difference shows the limit of multiplying two isolated g_m/g_{ds} stage estimates. The complete model includes loading between stages, body transconductance, frequency-dependent intrinsic elements, and the finite impedances of bias networks.

3.3 Input common-mode range

The ICMR testbench connects the amplifier as a unity-gain follower and sweeps the common-mode input. The tracking error is

$$e_{CM} = V_{OUT} - V_{CM}. \tag{3.6}$$

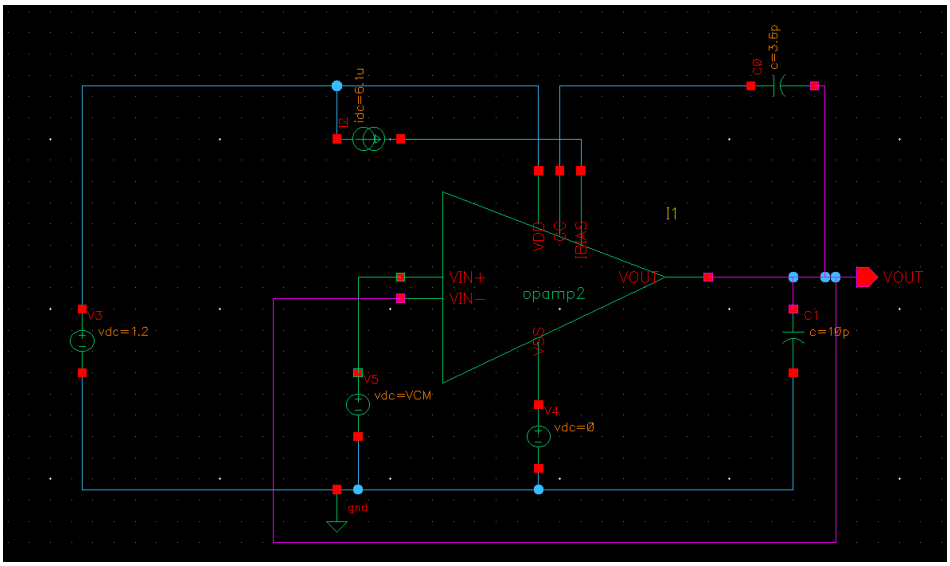


Figure 3.3: Unity-gain follower testbench used for the ICMR sweep.

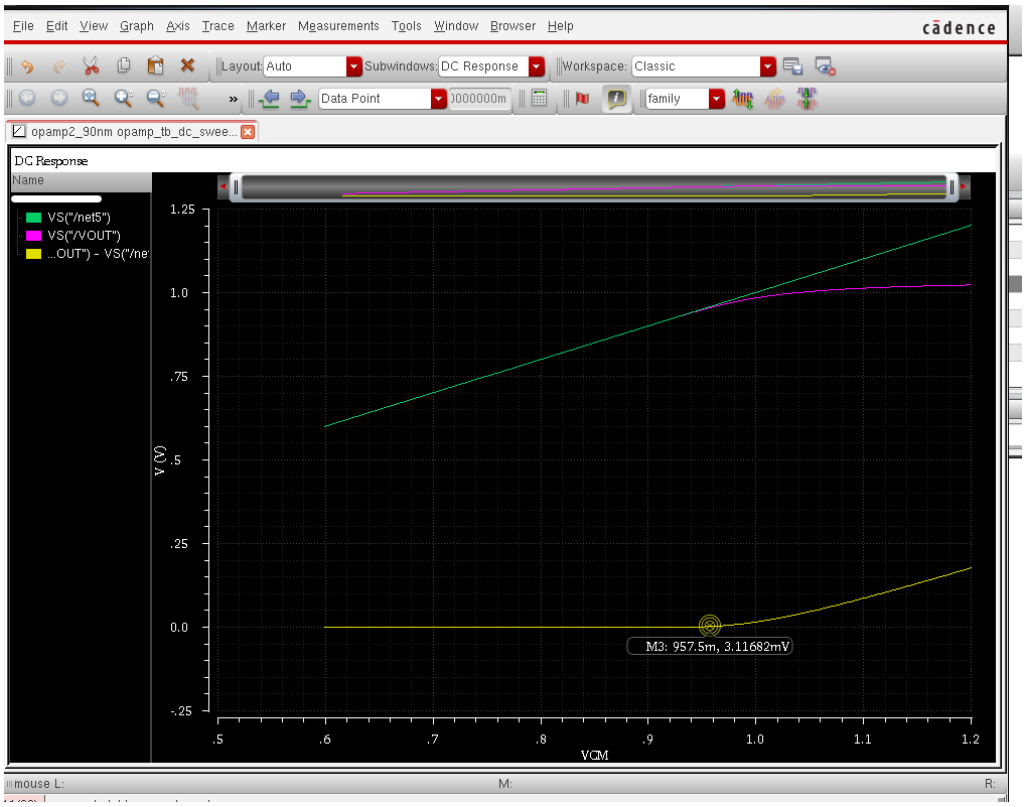


Figure 3.4: Upper ICMR sweep. The marked point is $V_{CM} = 0.9575\,\mathrm{V}$ with approximately $3.12\,\mathrm{mV}$ tracking error.

The upper boundary is therefore approximately $0.958\,\mathrm{V}$ under the shown error criterion, so the required $0.9\,\mathrm{V}$ upper limit is satisfied. The theoretical design boundary was $0.9\,\mathrm{V}$; the nominal transistor model provides some extra margin.

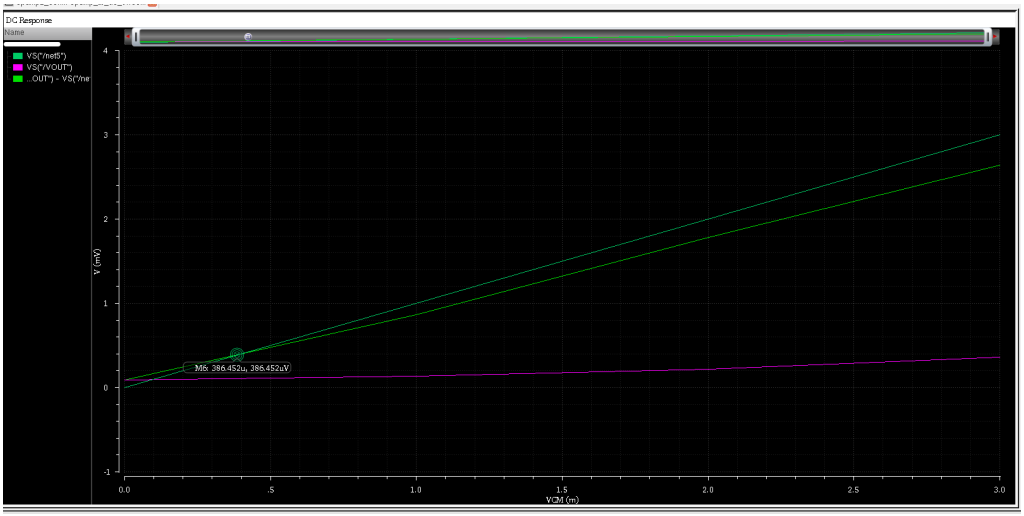


Figure 3.5: Available lower-bound plot. The horizontal axis is displayed in millivolts and spans only 0–3 mV; therefore this plot does not validate a 0.4 V lower ICMR limit.

The conservative square-law estimate is

$$V_{ICMR,min} \approx V_{SS} + V_{OV1} + V_{TN,max} + V_{DS5,sat} = 0 + 0.1592 + 0.1866 + 0.1042 \approx 0.450\,\text{V}. \quad (3.7)$$

Consequently, the lower ICMR is reported as *not yet signed off*. The dc sweep must be repeated over, for example, 0.2–0.6 V, not 0–3 mV. The target interval 0.4–0.9 V should only be marked as passed if the rerun keeps $|e_{CM}|$ below the chosen limit and all critical devices remain saturated.

3.4 Output-voltage swing

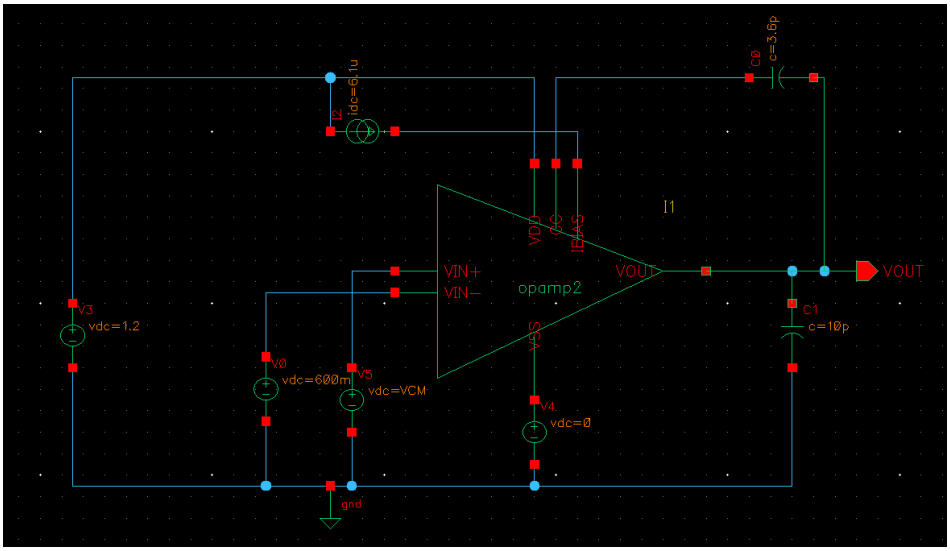


Figure 3.6: DC output-swing testbench. A small differential input sweep drives the open-loop output into both limits.

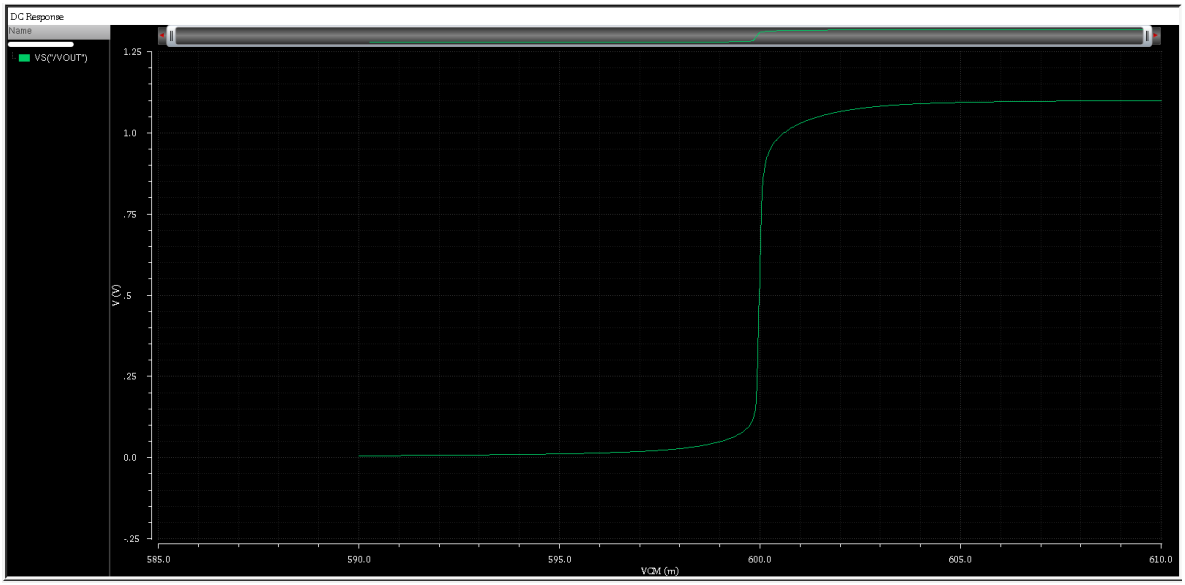


Figure 3.7: Output voltage versus differential input near the switching point.

From the plateaus, the no-load output swing is approximately

$$V_{OUT,min} \approx 0.01\,\mathrm{V}, \qquad V_{OUT,max} \approx 1.10\,\mathrm{V}, \qquad V_{OUT,pp} \approx 1.09\,\mathrm{V}. \tag{3.8}$$

The upper headroom is about 0.10 V below V_{DD} and the lower headroom is about 0.01 V above V_{SS}. This is a dc/no-load result; a resistive load or large output current will reduce the available swing.

Chapter 4

AC Analysis

4.1 Open-loop gain, bandwidth, and phase margin

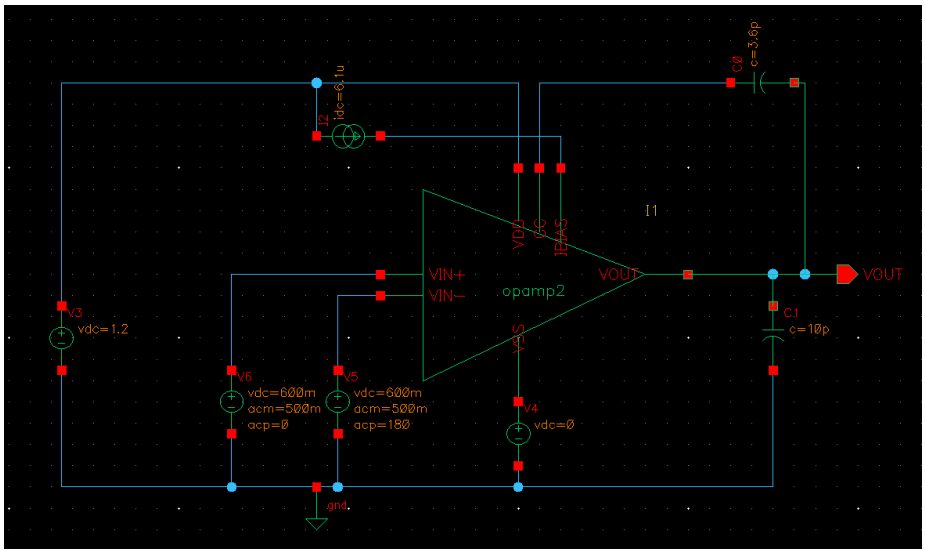


Figure 4.1: Open-loop differential ac testbench with $C_L = 10\,\text{pF}$ and $C_C = 3.6\,\text{pF}$.

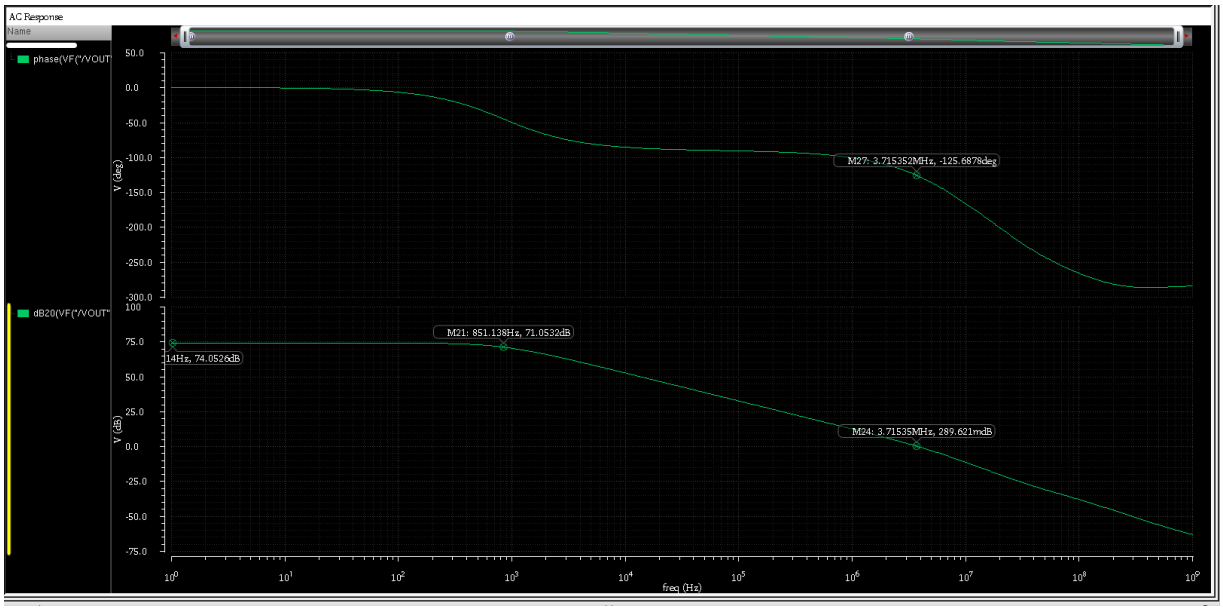


Figure 4.2: Open-loop magnitude and phase response.

The marked measurements are

$$A_0 = 74.0526\,\text{dB}, \tag{4.1}$$

$$f_{-3\,dB} = 851.138\,\text{Hz}, \tag{4.2}$$

$$|A(j2\pi f)| = 0.2896\,\text{dB at } f = 3.71535\,\text{MHz}, \tag{4.3}$$

$$\angle A(j2\pi f) = -125.688^\circ \text{ at } f = 3.71535\,\text{MHz}. \tag{4.4}$$

The actual 0-dB crossing is slightly above the marked frequency. Using the closest common marker gives

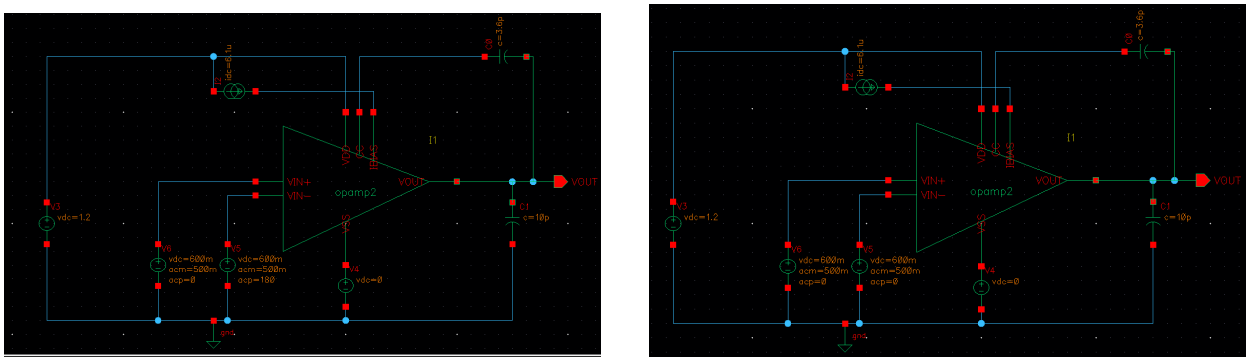
$$f_u \approx 3.72\,\text{MHz}, \qquad PM \approx 180^\circ - 125.688^\circ = \boxed{54.31^\circ}. \tag{4.5}$$

Thus the design is stable but has less margin than the 60° target. The simulated $g_{m6} \approx 551\,\mu\text{S}$ is lower than the 691 μS first-cut value, which explains part of the reduced pole separation. The right-half-plane zero created by a bare Miller capacitor also reduces phase margin near unity gain.

4.2 Common-mode rejection ratio

For differential excitation, $v_{in+} = +v_{ac}/2$ and $v_{in-} = -v_{ac}/2$. For common-mode excitation, both inputs receive the same ac signal. The low-frequency CMRR is

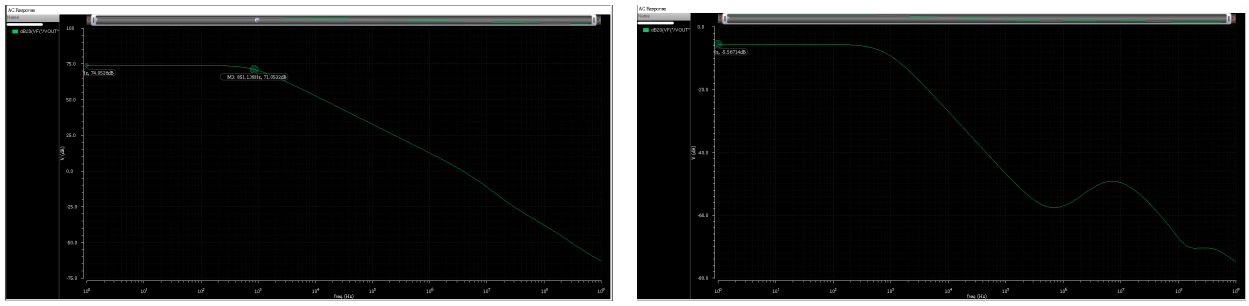
$$CMRR_{dB} = A_{d,dB} - A_{cm,dB}. \tag{4.6}$$



(a) Differential-gain testbench.

(b) Common-mode-gain testbench.

Figure 4.3: CMRR test configurations.



(a) $A_d(0) = 74.0526\,\text{dB}$.

(b) $A_{cm}(0) = -5.56714\,\text{dB}$.

Figure 4.4: Differential and common-mode ac responses.

Therefore,

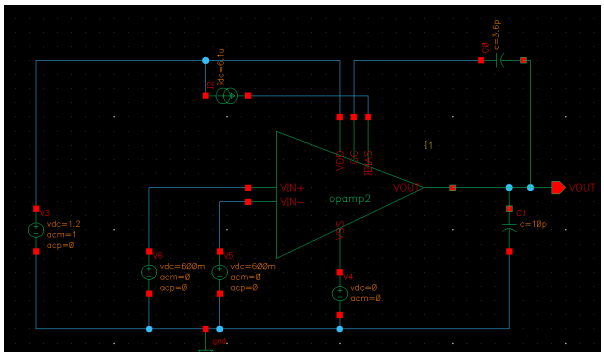
$$CMRR(0) = 74.0526 - (-5.56714) = \boxed{79.62\,\text{dB}}. \tag{4.7}$$

It is important not to label the 74.05 dB differential-gain plot itself as CMRR; both A_d and A_{cm} are required.

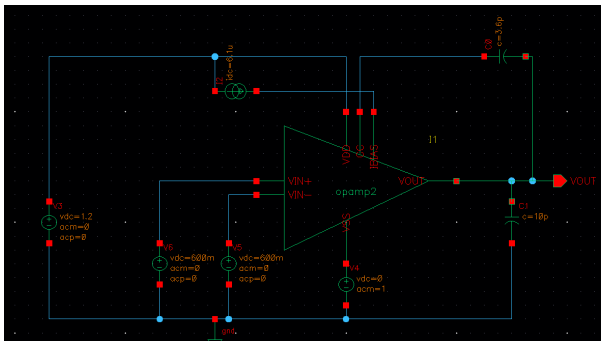
4.3 Power-supply rejection

For PSRR$^+$, a 1 V ac signal is applied to V_{DD} while the signal inputs remain at their dc common-mode level and V_{SS} is ac grounded. For PSRR$^-$, the injection is applied to V_{SS} while V_{DD} is ac grounded. With unity ac injection,

$$PSRR^{\pm}_{dB} = A_{d,dB} - 20\log_{10}\left|\frac{V_{OUT}}{V_{SUPPLY}}\right|. \tag{4.8}$$

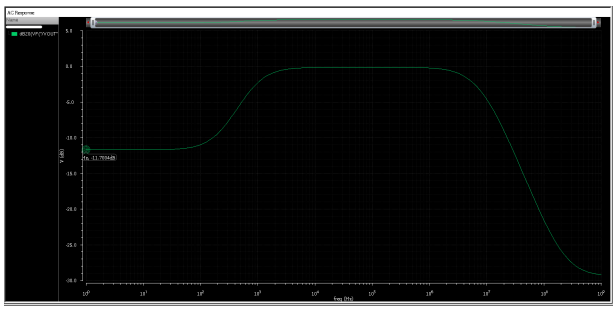


(a) PSRR$^+$ testbench.

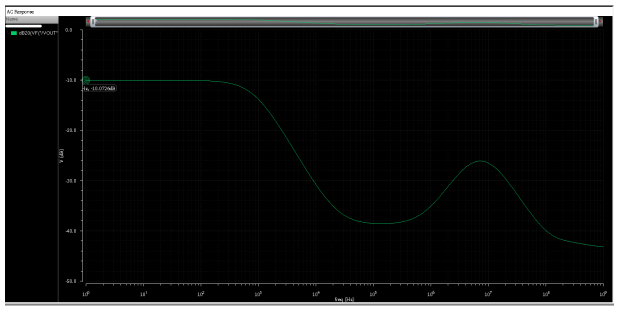


(b) PSRR$^-$ testbench.

Figure 4.5: Supply-ripple injection configurations.



(a) $A_{VDD}(0) = -11.7004\,\text{dB}$.



(b) $A_{VSS}(0) = -10.0726\,\text{dB}$.

Figure 4.6: Supply-to-output transfer functions. These curves are not PSRR until subtracted from differential gain.

The low-frequency rejection values are

$$PSRR^+(0) = 74.0526 - (-11.7004) = \boxed{85.75\,\text{dB}}, \tag{4.9}$$

$$PSRR^-(0) = 74.0526 - (-10.0726) = \boxed{84.13\,\text{dB}}. \tag{4.10}$$

The rejection degrades with frequency because supply disturbances couple through device capacitances and because the open-loop differential gain rolls off.

Chapter 5

Transient Analysis

5.1 Unity-gain step response

The amplifier is connected as a voltage follower and driven by a 0.4 V step between 0.4 V and 0.8 V. The step stays inside the intended ICMR and the load is 10 pF.

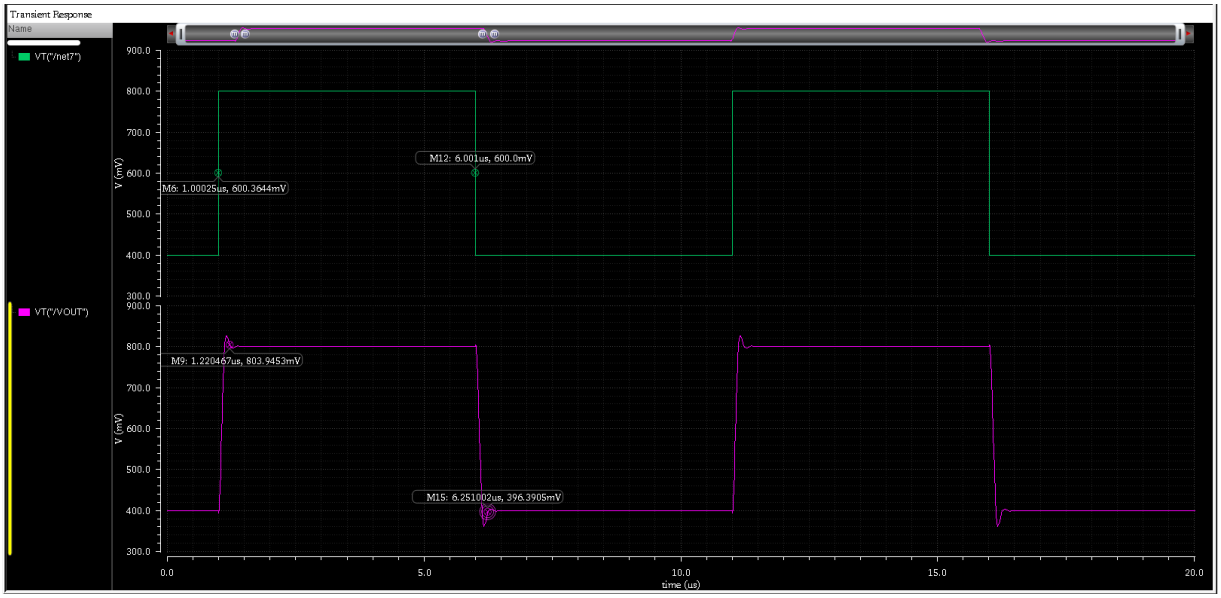


Figure 5.1: Input and output step responses used to evaluate overshoot and 1% settling time.

For a final value of 0.8 V, the 1% band is 0.796–0.804 V. For a final value of 0.4 V, it is 0.396–0.404 V. The markers show

$$t_{s,+} \approx 1.220467\,\mu\text{s} - 1.000250\,\mu\text{s} = \boxed{0.220\,\mu\text{s}}, \tag{5.1}$$

$$t_{s,-} \approx 6.251002\,\mu\text{s} - 6.001000\,\mu\text{s} = \boxed{0.250\,\mu\text{s}}. \tag{5.2}$$

Both are comfortably below the selected 1 μs requirement. The positive edge exhibits a small overshoot, while the negative edge has a larger undershoot; this asymmetry is consistent with different sourcing and sinking paths.

5.2 Slew rate

The analytical value is

$$SR_{calc} = \frac{I_5}{C_C} = \frac{18\,\mu\text{A}}{3.6\,\text{pF}} = \boxed{5\,\text{V}/\mu\text{s}}. \tag{5.3}$$

From the derivative plot supplied during simulation, the negative slew marker is approximately $-3.97\,\text{V}/\mu\text{s}$, while the positive edge is approximately $4.7\,\text{V}/\mu\text{s}$. These values are slightly below the ideal calculation because part of the available current charges internal capacitances and because the output stage is not an ideal current source. For final sign-off, the derivative plot should be saved with both positive and negative flat-slope markers visible.

Chapter 6

Results and Discussion

6.1 Specification summary

Table 6.1: Target versus nominal simulation.

Metric	Target	Result	Status / comment
DC gain	$\geq 70\,\mathrm{dB}$	$74.05\,\mathrm{dB}$	Pass; margin of $4.05\,\mathrm{dB}$
Unity-gain frequency	$5\,\mathrm{MHz}$	$\approx 3.72\,\mathrm{MHz}$	Not met; exact 0-dB marker should be added
Phase margin	$\geq 60^\circ$	$\approx 54.3^\circ$	Not met; stable but limited margin
Dominant-pole bandwidth	—	$851.14\,\mathrm{Hz}$	Informational
Slew rate	$5\,\mathrm{V}/\mu\mathrm{s}$	$+\approx 4.7,\ -\approx 3.97\ \mathrm{V}/\mu\mathrm{s}$	Not met; analytical value is $5\,\mathrm{V}/\mu\mathrm{s}$
1% settling time	$\leq 1\,\mu\mathrm{s}$	$0.220\,\mu\mathrm{s}/0.250\,\mu\mathrm{s}$	Pass
ICMR	0.4–$0.9\,\mathrm{V}$	upper $\approx 0.958\,\mathrm{V}$	Re-test; lower sweep has wrong mV range
Output swing	maximize	≈ 0.01–$1.10\,\mathrm{V}$	$1.09\,\mathrm{Vpp}$, no-load dc
CMRR, low frequency	—	$79.62\,\mathrm{dB}$	Calculated from $A_d - A_{cm}$
PSRR$^+$, low frequency	—	$85.75\,\mathrm{dB}$	Calculated from $A_d - A_{VDD}$
PSRR$^-$, low frequency	—	$84.13\,\mathrm{dB}$	Calculated from $A_d - A_{VSS}$
Quiescent output	near mid-supply	$0.598\,\mathrm{V}$	Pass
Estimated dc power	—	$154.5\,\mu\mathrm{W}$	Analytical estimate, not supply-current measurement

6.2 Main tradeoffs

The final circuit is functional and fast in closed loop, but the gain, unity-gain frequency, phase margin, and negative slew rate do not reach their nominal targets simultaneously. The main tradeoffs are:

- Reducing C_C raises GBW and slew rate, but generally reduces phase margin.
- Increasing g_{m1} raises GBW but requires more input-stage current or larger M1–M2 devices, increasing power or capacitance.
- Increasing g_{m6} improves non-dominant-pole separation, but raises second-stage current and may worsen the RHP-zero effect.
- A series nulling resistor $R_z \approx 1/g_{m6}$ can remove or move the RHP zero. With $g_{m6} \approx 551\,\mu\text{S}$, $1/g_{m6} \approx 1.81\,\text{k}\Omega$. This can recover phase margin while allowing a smaller C_C, but must be re-simulated across corners.
- Longer channel lengths can increase intrinsic gain by reducing g_{ds}, but normally reduce speed and increase area.

6.3 Required final checks

Before tape-out-quality sign-off, the following simulations should be added:

1. repeat the lower ICMR sweep over 0.2–0.6 V;
2. place an exact 0-dB marker and read phase at that same frequency;
3. save positive and negative derivative markers for slew rate;
4. measure total supply current directly and report $P_{DC} = V_{DD}I_{DD}$;
5. extract output resistance using a small-signal current injection at the output;
6. run input-referred noise analysis, because noise is not present in the available result set;
7. perform PVT and Monte Carlo mismatch analysis, particularly for gain, ICMR, output quiescent voltage, CMRR, and phase margin.

Chapter 7

Conclusion

An eight-transistor, two-stage Miller-compensated CMOS operational amplifier was designed in GPDK090 for a 1.2 V supply and a 10 pF load. The calculation flow produced a consistent initial bias and sizing set, including $I_5 = 18\,\mu$A, $S_5 = 11.18$, and $S_8 = 3.79$. The dc operating point confirms balanced differential currents, balanced output-stage currents, saturation-region operation, and a quiescent output close to mid-supply.

The nominal simulation demonstrates useful performance: 74.05 dB dc gain, approximately 3.72 MHz unity-gain frequency, approximately 54.3° phase margin, sub-0.3 μs 1% settling, about 1.09 Vpp no-load output swing, 79.62 dB CMRR, and more than 84 dB low-frequency PSRR. The revised dc-gain requirement of at least 70 dB is satisfied with 4.05 dB margin. The 5 MHz, 60°, and 5 V/μs targets are not all met. The most direct next iteration is to increase second-stage transconductance and consider a Miller nulling resistor so that GBW can be raised without sacrificing phase margin. The lower ICMR must also be re-simulated with the correct voltage sweep before it is claimed as a verified result.

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[3] P. E. Allen, *CMOS Analog Circuit Design: Design Procedure for the Unbuffered Two-Stage CMOS Operational Amplifier*, course lecture material, 2016.

[4] Cadence Design Systems, *Virtuoso Analog Design Environment and Spectre Circuit Simulator*, simulation environment used for this project.

[5] GPDK090 BSIM3 MOSFET models, nominal-corner transistor models used in the schematic simulations.

Appendix A

Measurement Definitions

$$A_{d,dB}(f) = 20\log_{10}\left|\frac{V_{OUT}}{V_{ID}}\right|, \tag{A.1}$$

$$A_{cm,dB}(f) = 20\log_{10}\left|\frac{V_{OUT}}{V_{ICM}}\right|, \tag{A.2}$$

$$CMRR_{dB}(f) = A_{d,dB}(f) - A_{cm,dB}(f), \tag{A.3}$$

$$PSRR^{+}_{dB}(f) = A_{d,dB}(f) - 20\log_{10}\left|\frac{V_{OUT}}{V_{DD,ac}}\right|, \tag{A.4}$$

$$PSRR^{-}_{dB}(f) = A_{d,dB}(f) - 20\log_{10}\left|\frac{V_{OUT}}{V_{SS,ac}}\right|, \tag{A.5}$$

$$SR_{+} = \max\left(\frac{\mathrm{d}V_{OUT}}{\mathrm{d}t}\right), \qquad SR_{-} = \min\left(\frac{\mathrm{d}V_{OUT}}{\mathrm{d}t}\right). \tag{A.6}$$

For settling time, the start time is the input's 50% crossing and the stop time is the final entry into the specified error band without any later excursion.